# Equal Pixels, Unequal Views: Field of View Limits Robust AI-Generated Image Detection

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## Abstract

Fixed-size crops at native resolution preserve local image traces, but their field of view shrinks as image size increases. We study how this preprocessing choice affects AI-generated image detection under degradation, distinguishing each crop's field of view from the total image area covered by multiple crops. With frozen, crop-trained DINOv2 detectors, replacing a central crop with a resized whole image raises robust ROC AUC from 0.886 to 0.963 on the NTIRE 2026 hard validation split used for policy selection. On large held-out SocialRF images with synthetic degradations, whole-image input improves AUC by 7.9–9.1 percentage points. Factorial experiments show that downscaling before cropping benefits DINOv2, CLIP, and B-Free on this benchmark, while increasing coverage at fixed scale provides little additional gain. Resizing controls support a role for wider per-crop context, although interpolation and changes in crop count limit causal attribution. A separately trained whole-image system improves NTIRE test robust AUC by 0.062 while reducing clean AUC by 0.014. Benefits therefore depend on the detector, dataset, and degradation. We additionally identify format and size confounds in two evaluated benchmark configurations. These findings motivate reporting crop field of view and resize factors alongside detector input resolution.


## 1 Introduction

Deployed detectors of AI-generated images see what social platforms deliver: images that have been recompressed, resized and filtered, at every size. A detector must turn each one into a fixed-size input. Forensic work argues that this step should not resize: "Most of all, it is important to avoid resizing" because interpolation "may erase the subtle high-frequency traces left by the generation process" (Corvi et al. 2023). The usual answer is a fixed-size crop at native resolution, which we call a *window*, scored once or several times with the scores averaged (Corvi et al. 2023; Li et al. 2025a; Guillaro et al. 2025; Konstantinidou, Koutlis, and Papadopoulos 2025).

A fixed input resolution does not fix how much of the source image a detector observes. We call each square input a *view*, and define a crop's field of view as its side length $s$ divided by the source image's short side $h$: $\mathrm{FOV} = s/h$. A 200-pixel crop spans 39% of a 512-pixel image's short side but only 5% of a 4000-pixel image's. Scoring additional native crops increases total image *coverage* while retaining this narrow field of view in each input. Downscaling before cropping instead widens each input's field of view, while also changing resampling and aggregation. Which aspects of input presentation account for robustness gains on degraded images?

We investigate this question using detectors trained for the NTIRE 2026 Robust AI-Generated Image Detection challenge (Gushchin et al. 2026). All 407k training images have a short side of at most 512 pixels. Augmentation changes the field of view, but no sampled training view spans less than approximately 20% of the short side. On the hard validation split, a native window can span only 5%, and robust AUC falls as image size grows. Presenting the whole image to the frozen crop-trained models improves robust AUC by 7 to 9 points per branch. We examine this effect through controlled changes to coverage and downscale factor, and evaluate its transfer and clean-image cost on held-out data.

**Contributions.**

- **Preprocessing diagnosis.** Whole-image presentation substantially improves robust AUC for frozen, crop-trained DINOv2 detectors, exceeding the gain from averaging nine native crops while requiring only one view (Sec. 4).
- **Controlled analysis.** We vary coverage and downscale factor independently within a tiled protocol and use resizing controls to examine competing explanations. On degraded SocialRF, downscaling benefits DINOv2, CLIP, and B-Free, while additional coverage contributes little (Secs. 5, 6, and 8).
- **Transfer and limitations.** We report external evaluations, a declared second NTIRE test evaluation, clean-image losses, and unsuccessful comparisons. We also identify format and size confounds in the evaluated CommunityAI and Synthbuster/RAISE configurations (Secs. 8 and 9).

Whole-image resizing already appears in detection pipelines. The NTIRE 2026 winner resizes at inference to preserve global spatial context, the fifth-placed team uses *squish* (Gushchin et al. 2026), and AI-GenBench (Pellegrini

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et al. 2025) finds resizing often preferable to multi-crop evaluation for CLIP and DINOv2. We investigate the explanations behind these observations by varying per-crop field of view and total coverage, and by testing resizing controls. The contribution is an empirical analysis of when these choices help and where they fail.

## 2 Related work

**Crop, don't resize.** Corvi et al. (2023) recommend training on native patches cropped without resizing and deciding on the whole image by fusion. SAFE (Li et al. 2025a) replaces down-sampling with a crop and reports that the resize "indeed diminishes the subtle artifact features". B-Free (Guillaro et al. 2025) trains on 504-pixel crops and averages native-resolution crops at test time. SPAI (Karageorgiou et al. 2025) learns a spectral model so that any resolution can be scored without resizing. TextureCrop (Konstantinidou, Koutlis, and Papadopoulos 2025) selects textured native crops and notes that preprocessing loss grows with native resolution. HiDA-Net (Mu et al. 2026) tiles the full image and reports 84.2% accuracy from its global view alone against 96.1% with tiles. All of these reason about high-frequency traces. SAFE's own ablation is useful context for us: its CNN scores a centre crop and the full native image about equally in that setting; field-of-view sensitivity therefore need not transfer to every detector.

**Resize can win.** AI-GenBench (Pellegrini et al. 2025) reports resize at least as good as five-crop for fine-tuned DINOv2 ViT-L/14 (94.24 against 93.44 AUROC) and clearly better for CLIP (92.04 against 87.66). In AIGIBench's test-time comparison (Li et al. 2025b), the one CLIP-based detector does better with resize while all five low-level detectors do better with crops. Cozzolino et al. (2024) resize the short side before a centre crop.[1] WaRPAD (Choi, Lee, and Lee 2025) rescales the whole image to a fixed square before tiling it, which holds each tile's field of view constant, although the paper motivates it differently. GlobalForge (Cui et al. 2026) reports the opposite of our test-time result on clean data: a UnivFD probe trained with centre crops loses about 9 balanced-accuracy points (67.9 to 58.7, their Fig. 2) when tested with resize. We return to this in Sec. 8.

**Scale and size as confounds.** Grommelt et al. (2025) show that JPEG quality and image size separate real from fake in GenImage, and note in one sentence that "detectors employing cropping to achieve a uniform input size are still subject to bias in terms of object size". Their concern is a size shortcut correlated with the label. In our training set, native size has a near-chance monotone size-label AUC, yet performance varies with size at test time. Rajan et al. (2025) show that random resized crops inject resampling signals when real and fake sizes differ, and BIAS-ID (Ricker, Fischer, and Quiring 2026) measures detector score shifts under such transformations. FixRes (Touvron et al. 2019) provides a related account of train-test differences in apparent object scale. We examine a possible mismatch in per-crop field of view, while retaining content scale and resampling as alternative explanations (Sec. 6).

[1] In their released code, https://github.com/grip-unina/ClipBased-SyntheticImageDetection.

## 3 Setup

**Data.** We train on 407,223 images: the NTIRE 2026 training shards, COCO and Open Images reals, and paired real and generated images from several open and commercial generators. Every training image has a short side of 200 to 512 pixels (median 512, and 384 or more for 95% of images). We evaluate on the organisers' labelled splits: `ntire_val` (10,000 images, half distorted), `ntire_val_hard` (2,500, half distorted) and the public test split `ntire_test` (2,500). Robust AUC is the ROC AUC over distorted images, and clean AUC is over undistorted ones, following the challenge (Gushchin et al. 2026).

**Models.** Two branches built on DINOv2-L/14 with registers (Oquab et al. 2024; Darcet et al. 2024), all 24 blocks fine-tuned, mean-pooled patch tokens, 0.91B parameters in total. Branch A is one tower at a 512-pixel input, trained for 2 epochs. Branch B concatenates two towers at 224 pixels, trained for 3 epochs. Each source image yields one clean view and ten degraded views drawn from a pool of 26 operations in six families, chained up to five deep, including a neural codec (Ballé et al. 2018) and a downscale applied before the window is taken. The loss is class-balanced cross-entropy plus a degradation-prediction term (weight 0.3) and clean-degraded consistency terms on the outputs (KL) and hidden features (MSE), each at unit weight. The schedule is cosine with 3% warm-up, with weight averaging (Izmailov et al. 2018) over the last quarter. On one RTX 4090 the squish-trained branches took about 19 hours (B) and 29 hours (A).

**Views, field of view and coverage.** A *view* is one square input to the model: a region of the image resized to the branch's input size. A detector may score several views of one image and average their logits. The *field of view* of a view is the share of the image's short side it shows. *Coverage* is the share of the image area that all views of one image show together. A 200-pixel window cut after the image is downscaled by a factor $k$ has $\mathrm{FOV} = 200k/h$, provided the window fits. For a fixed-size window, downscaling widens its field of view and increases the source area it covers. In our fully tiled construction below, total coverage is held fixed while the downscale factor varies.

The *crop* policy takes one 200-pixel window at native resolution (the centre window at test time) and resizes it to the branch's input size. Images with a short side under 200 pixels are taken whole. The *squish* policy resizes the whole image to the input size, discarding aspect ratio: area interpolation on the short side down to the input size when the image is larger, then area interpolation to the square. Both policies give every image the same number of input pixels. Under squish, field of view and coverage are both 1. To vary the two one at a time, *tiles*$(k, f)$ takes the centred region holding a fraction $f$ of the image area, downscales it by $k$ with area interpolation, covers it with 200-pixel windows, resizes each window to the input size and averages the logits. $f$ sets coverage and

| split | n | p5 | p50 | p95 |
|---|---|---|---|---|
| train, native | 407,223 | 0.391 | 0.391 | 0.521 |
| train, window taken | 220,000 | 0.391 | 0.468 | 1.000 |
| train, after second crop | 220,000 | 0.296 | 0.415 | 0.942 |
| `ntire_val` | 10,000 | 0.100 | 0.312 | 0.893 |
| `ntire_val_hard` | 2,500 | 0.112 | 0.298 | 0.784 |

Table 1: Field of view of the 200-pixel window (200/short side). *Window taken* is the window as the trainer cuts it, on 20,000 seeded training images with 11 views each. A downscale before the window widens the field of view: 24.9% of views exceed 0.521 and 6.3% show the whole short side. A crop after the window narrows 18% of views below 0.391, to no less than 0.195, and enlarges them back. On `ntire_val_hard`, 70.8% of images fall below 0.391, and the smallest views (0.05) are about 4 times narrower than any training view.

$k$ sets each view's field of view. Downscaling also changes interpolation and the number of windows averaged, so these are experimental contrasts rather than isolated causal effects.

**Comparison types.** Frozen-model comparisons change presentation at test time while retaining branch checkpoints. Training comparisons use separately crop-trained and squish-trained models. The final test read compares complete systems, including their checkpoints and fitted fusion settings. We distinguish these throughout.

**Fusion and selection.** The two branches' logits are z-scored and combined as $wz_A + (1-w)z_B$. The z-score parameters and the weight $w$ are fitted on `ntire_val` alone, by the mean of robust and clean AUC. For the squish models the choice between final and averaged weights is made the same way. For the crop models we kept the choice made earlier on both validation splits. The fitted weights were $w = 0.50$ for crop and $w = 0.55$ for squish.

**Statistics.** All intervals are 95% percentile intervals from 2,000 paired bootstrap resamples, stratified by label and distortion status, with the same resample indices for every model being compared.

**Protocol.** The test split was first read with the crop fusion (2026-10-04). Before training any squish model, we committed a registration of the squish study that allowed one further test read. Registered experiments and exploratory analyses are distinguished throughout. Amendments were recorded before the corresponding results existed; the supplement lists their outcomes, including failed or unidentifiable tests. The later diagnostic experiments did not evaluate the test split again. We chose squish after observing its effect on `ntire_val_hard`, which we label the *selection split*. External evaluations and the declared second test read provide evidence beyond policy selection. Code, registrations, and per-image scores will be released with the paper.

## 4 Equal pixels, unequal views

Table 1 shows how much of each image the window covers. The central 90% of native training field-of-view values lie between 39 and 52% of the short side. Augmentation widens some views and narrows others; the minimum observed after the second crop is approximately 20%. On `ntire_val_hard` the median is 30%, and one image in twenty gets 11% or less. Under the crop policy, fused robust AUC on `ntire_val_hard` falls from 0.911 for images with a short side of 80 to 336 pixels to 0.865 for images above 752 pixels. Figure 1 shows the same fall in eight equal-count bins.

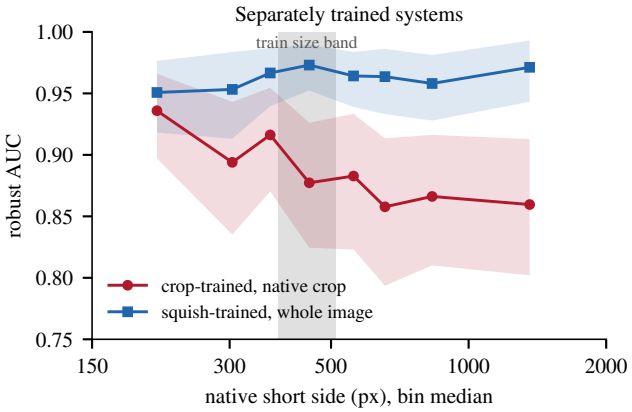



Figure 1: Fused robust AUC on the distorted rows of `ntire_val_hard` (selection split) in eight equal-count bins of native short side, with 95% bootstrap bands. Red: the crop-trained fusion under its 200-pixel window. Blue: the separately squish-trained fusion. This is a comparison of trained systems, not the frozen-checkpoint intervention in Table 2. Grey: the 384–512-pixel band containing at least 95% of native training short sides.

| branch | presentation | views | robust | clean |
|---|---|---|---|---|
| A (512) | centre window | 1 | 0.8561 | 0.9912 |
| | grid9 | 8.9 | 0.8495 | 0.9978 |
| | fov3 | 8.9 | 0.8703 | 0.9977 |
| | squish | 1 | **0.9482** | 0.9906 |
| B (224) | centre window | 1 | 0.8872 | 0.9801 |
| | grid9 | 8.9 | 0.9049 | 0.9931 |
| | fov3 | 8.9 | 0.9215 | 0.9942 |
| | squish | 1 | **0.9599** | 0.9886 |

Table 2: Frozen crop-trained branches under four test-time presentations on the hard validation split (selection split). Averaging additional native windows (grid9) helps B by 1.8 points and costs A 0.7. Whole-image input gives the largest robust AUC for both branches.

To test whether additional native crops recover the gain, we scored the two frozen crop-trained branches under four test-time presentations (Table 2). *grid9* averages nine 200-pixel native windows, nine times the pixels at the same field of view. *fov3* first downscales the short side to 600 pixels and then takes the nine windows, which sets the field of view to $1/3$. *squish* shows the whole image once.

Additional native windows yield smaller changes than

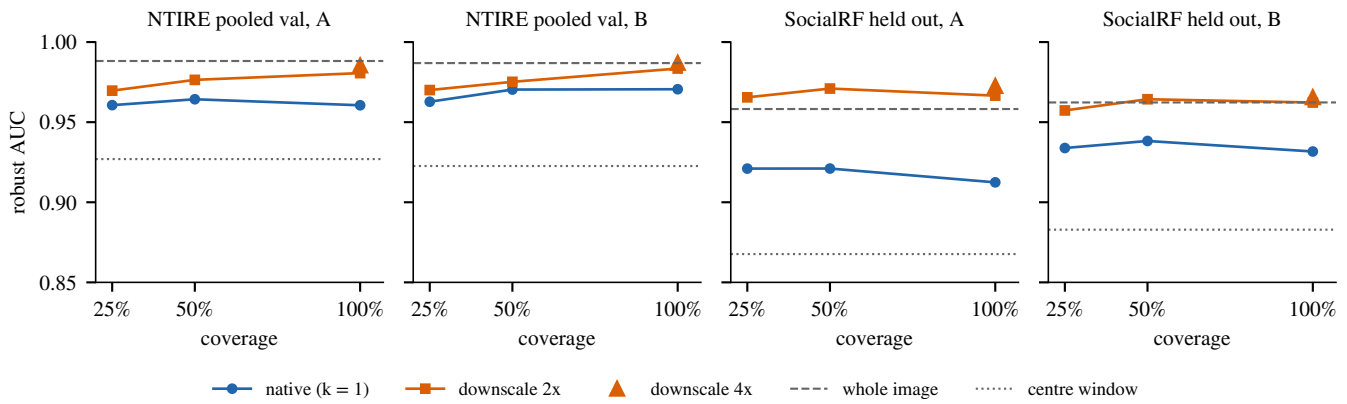



Figure 2: Robust AUC of *tiles*$(k, f)$ for the frozen crop-trained branches on images with a short side of at least 800 pixels. Moving right tiles more of the image with same-size windows (coverage). Different lines vary the downscale factor, also changing field of view and window count. NTIRE pools both validation splits. On held-out SocialRF, coverage contributes little additional gain under our degradation chains. Dashed: squish. Dotted: the centre window.

whole-image input: grid9 gains 1.8 robust points on B and loses 0.7 on A. The intermediate fov3 presentation gains 1.4 and 3.4 points, and squish gains 9.2 and 7.3. Squish uses one view and the same pixel count as the centre window. On B the paired interval for the squish gain is [+0.057, +0.090]. With z-scores and weight fitted on `ntire_val`, the fused robust AUC rises from 0.886 to 0.963 and the true positive rate at 1% false positive rate from 0.516 to 0.752. Clean AUC moves by $-0.001$ on A and $+0.009$ on B.

Squish changes three things at once. It raises coverage to the whole image, widens the field of view of its one view, and resamples local image traces and degradations. Figure 2 previews the coverage/downscale contrasts of Sec. 6, following controls for resolution cues and aspect ratio.

## 5 Controls for resolution and aspect ratio

Resizing can leave cues about native resolution: small images are enlarged and large images reduced. Four controls test whether such cues explain the gain. First, native short side has a monotone size-label AUC of 0.528 in training. This weakens a simple size-shortcut account but does not exclude nonlinear cues. Second, within native-size deciles, where resize factors vary less, squish beats the centre window in all ten deciles on branch B (size-weighted robust AUC 0.888 to 0.964). Third, band-limiting the squished input to 160 pixels reduces available resolution cues, yet B retains +7.2 of its +7.3 points and A retains +5.7 of +9.2. This control also removes image detail and cannot guarantee that all size cues disappear. Fourth, on rows with only photometric transforms (median short side 784 pixels), the gain is +9.0 points on A and +8.3 on B, with intervals above zero. The effect therefore does not require the challenge's geometric transforms (Gushchin et al. 2026).

Squish also stretches non-square images, potentially introducing frequency artefacts along one axis (Grommelt et al. 2025). Two aspect-preserving presentations, a centre square after resizing the short side and a padded letterbox, retain 93 to 100% of its robust gain. Aspect-ratio distortion is therefore unnecessary for most of the measured improvement.

## 6 Coverage and downscale contrasts

This section uses frozen crop-trained checkpoints throughout. The NTIRE analyses pool both validation splits; SocialRF provides external data with our synthetic degradation chains.

**Degradation and size.** We took the 6,250 undistorted images of the two validation splits and applied JPEG (quality 90, 70, 30), Gaussian blur ($\sigma$ 0.5, 1, 2) or Gaussian noise ($\sigma$ 0.02, 0.05, 0.10) at native resolution. The robust gain of squish over the centre window, averaged over the three operations, rises from $-0.001$ with no degradation to $+0.029$ at the heaviest level on branch A, and from $+0.004$ to $+0.027$ on B. Every rise from none to heavy has an interval above zero, pooled and in every size bucket. The rise is largest above 752 pixels ($+0.049$ on both branches). On undistorted images up to 736 pixels the two policies differ by at most 0.002. The levels are not matched in strength across operations, so we do not rank the operations.

**Coverage against downscale.** The images used here have a short side of at least 800 pixels. We scored *tiles*$(k, f)$ at $k \in \{1, 2\}$ and $f \in \{0.25, 0.5, 1\}$, so every window is a 200-pixel window like those in training. The coverage effect $d_F$ is the gain from $f = 0.25$ to $f = 1$ averaged over $k$. The downscale effect $d_K$ is the gain from $k = 1$ to $k = 2$ averaged over $f$. On NTIRE (4,068 images, 1,235 distorted, pooled validation splits) both effects are above zero on both branches (Table 3). On large degraded SocialRF images, which no model was trained or selected on, increasing coverage provides little additional gain: at a fixed downscale, showing the whole image instead of its central quarter changes robust AUC by less than 0.01. The downscale helps by 2.3 to 5.5 points at every coverage level, more than on NTIRE. Covering the whole image after a 4x downscale, *tiles*$(4, 1)$, is within 0.003 of squish on NTIRE and at least as good on SocialRF (A 0.973 against 0.958, B 0.966 against 0.962).

Figure 2 shows the stronger downscale effect on SocialRF, where the coverage curves are comparatively flat.

**Round-trip resizing control.** Round-trip resizing tests whether downscaling helps when each crop retains its original field of view. We downscale each region by two, restore its dimensions, and evaluate the original crop grid. Let $d_A$ be the AUC difference between this round trip and *tiles*$(1, f)$, averaged over $f$, and let $d_V = d_K - d_A$ be the residual contrast against *tiles*$(2, f)$. These are the *round-trip resize effect* and *residual downscale effect*, respectively. For DINOv2, $d_A$ is at most approximately 0.004 in magnitude on both datasets and branches, with intervals including zero; the residual contrast is positive (Table 3). This pattern is consistent with a benefit from wider per-crop context, but does not establish equivalence or identify a pure field-of-view effect. The round trip adds interpolation, and directly downscaled inputs average about four times fewer windows. These changes can affect both detail retention and pooling.

**Stronger downscales.** Squish downscales large images 5 to 18 times, far more than 2. To look at stronger factors we took 2,833 undistorted NTIRE images with a short side of at least 800 pixels and applied heavy JPEG (quality 30), blur ($\sigma = 2$) or noise ($\sigma = 0.10$) either before or after downscaling by $k$, then scored *tiles*$(k, 1)$. $R(k)$ is robust AUC with the degradation applied before the downscale minus after it, so a positive $R$ favors applying the degradation before downscaling. On branch A, $R(2) = +0.022$ [+0.019, +0.026] and $R(4) = +0.036$. On B they are +0.007 [+0.005, +0.008] and +0.013. $R(4)$ exceeds $R(2)$ on both branches, and noise, the degradation a downscale should average away most easily, gives the largest $R$ on both (A: +0.030 at 2x, +0.047 at 4x). This fits a downscale that also reduces the degradation, more so at stronger factors. It is not proof. The same degradation may hurt more when applied to a smaller image, JPEG blocks shrink with the image, and the round-trip control uses a different contrast. The ordering result is consistent with attenuation, but does not isolate it from changes in the effective scale of degradation.

**Possible mismatch in each view.** No sampled training view spans less than approximately 20% of the short side, while a 200-pixel crop of a 3,000-pixel photo spans 7%. Downscaling moves its field of view toward the observed training range, consistent with a train-test mismatch related to the scale discrepancy studied by FixRes (Touvron et al. 2019). However, the effect is not confined to images outside the native training size range: for short sides of 416–496 pixels, the separately trained fusions score 0.877 under crop and 0.973 under squish (Fig. 1). Content scale and resampling also vary, so the observed range alone does not establish the cause.

**CLIP.** We repeat the contrasts on three short crop-trained CLIP ViT-L/14 runs (Sec. 7) using the same SocialRF images. Downscaling improves robust AUC by 5.4 to 7.6 points, while increasing coverage provides little additional gain. The residual contrast $d_V$ is +0.041 to +0.063. The round-trip effect $d_A$ is also positive: +0.011 to +0.014 on degraded images, with intervals above zero on all seeds. However, it improves clean AUC by +0.022 to +0.025 as well. The round-trip benefit therefore cannot be attributed specifically to degradation removal.

| data | branch | $d_F$ | $d_K$ | $d_A$ | $d_V$ | $R(4)$ |
|---|---|---|---|---|---|---|
| NTIRE | A | +0.005* | +0.014* | −0.002 | +0.016* | +0.036* |
| | B | +0.011* | +0.008* | −0.001 | +0.010* | +0.013* |
| SocialRF | A | −0.004 | +0.050* | +0.001 | +0.049* | n/a |
| | B | +0.001 | +0.027* | +0.004 | +0.023* | n/a |

Table 3: Robust-AUC contrasts for frozen crop-trained branches, images with a short side of at least 800 pixels. $d_F$: coverage, from 25% to 100% of the image. $d_K$: downscale by 2. $d_A$: round-trip resize effect. $d_V$: residual downscale effect ($d_A + d_V = d_K$); neither is an isolated causal component. $R(4)$: robust AUC with heavy degradation applied before a 4x downscale minus after it, on 2,833 undistorted NTIRE images. NTIRE rows pool both validation splits. SocialRF rows use our degradation chains. * marks a 95% interval that excludes zero.

## 7 Training and baselines

**Training under whole-image input.** In short training runs (one tower, one epoch on a quarter of the training set, three seeds), squish-trained models beat crop-trained ones by +0.083 [+0.062, +0.105], +0.093 [+0.072, +0.115] and +0.078 [+0.057, +0.102] robust AUC on `ntire_val_hard`, with clean AUC within noise. We then trained the full branches under squish (Table 4). Their robust AUC is close to that of the frozen test-time intervention (A: 0.943 against 0.948, B: 0.959 against 0.960), with slightly higher clean AUC. The fused squish-trained models reach 0.960 robust and 0.995 clean, against 0.886 and 0.990 for the crop fusion: +0.074 [+0.058, +0.091]. The size gradient is gone: across the eight bins of Figure 1 the squish fusion stays between 0.951 and 0.973. In the short runs, training under squish adds 4 to 6 points beyond the test-time change, with DINOv2 and with CLIP (an exploratory comparison). The full branches show no such gap. They differ from the short runs in training length, number of views per image and loss terms, and we cannot say which of these closes the gap.

**A second backbone.** We repeated the short-run comparison with a CLIP ViT-L/14 tower (Radford et al. 2021) in place of DINOv2, every other setting unchanged. Over three seeds, training under squish beat crop by +0.139 to +0.150 robust AUC, with clean AUC near the crop baseline, and showing the crop-trained checkpoints the whole image gained +0.094 to +0.111 at test time alone, every interval clear of zero. Both gains grow with size: at test time, +0.168 above 752 pixels against +0.057 below 336. CLIP's crop-trained short runs are weaker than DINOv2's (0.748 robust against 0.799) and gain more.

**Is the crop baseline fair?** The crop policy enlarges its 200-pixel window to the input size, 1.12 times for B and 2.59 times for A, and averaging native windows may dilute one informative window. We checked both. A 224-pixel native

| branch | model | robust | clean |
|---|---|---|---|
| A | crop-trained | 0.8561 | 0.9913 |
| | crop-trained, squish at test | 0.9482 | 0.9906 |
| | squish-trained | 0.9425 | 0.9934 |
| B | crop-trained | 0.8873 | 0.9801 |
| | crop-trained, squish at test | 0.9599 | 0.9886 |
| | squish-trained | 0.9589 | 0.9898 |

Table 4: Per branch on `ntire_val_hard` (selection split). Training under squish yields similar point estimates to presenting the whole image to the frozen crop-trained model; this is not an equivalence test. Paired gains over crop-trained: A +0.092 and +0.086, B +0.073 and +0.072, all intervals clear of zero. The crop-trained rows were rescored for this table and differ from Table 2 by at most 0.0001.

window fed at 224, with no enlargement, scores within 0.013 of the 200-pixel window, and the squish-trained short runs beat it by +0.077 on average over three seeds. At 512 pixels a native 512-pixel window beats the enlarged 200-pixel one by +0.042 [+0.025, +0.058] at one seed, but it also has 2.56 times the field of view. A 196-pixel window fed at 196 pixels approximately matches the 200-pixel window's field of view without enlargement, providing a further contrast. On 630 large degraded images the native 512-pixel window leads by 0.059. The contrast between native 512- and 196-pixel inputs is +0.048 [+0.022, +0.072], while the 196-pixel input gains +0.011 [$-0.014$, +0.037], too uncertain to call, so whether enlargement handicaps branch A's crop baseline remains open. Multi-scale training with random resized crops (area 0.20 to 1.00) closes 73 to 81% of the gap between crop and squish, and squish stays ahead by +0.019 on average, every interval clear of zero. On the full frozen branches, a learned attention combiner over the nine native windows, fitted on `ntire_val`, gains about 3 points over averaging them. Squish, which is not fitted, still beats it by +0.069 [+0.052, +0.087] on A and +0.023 [+0.011, +0.036] on B, and ties it on B for images below 336 pixels.

## 8 Held-out evidence and where it stops

**External benchmarks.** We compare separately trained crop and squish fusions on three external benchmarks that no model was trained or selected on, after removing any image whose content digest matched an NTIRE image: AIGIBench SocialRF and CommunityAI (Li et al. 2025b), and Synthbuster with RAISE-1k reals (Bammey 2024; Dang-Nguyen et al. 2015). All three have no added degradation in their distributed form, so we report pooled AUC on the files as distributed, on every file re-encoded as JPEG at quality 95, and on every file passed through one seeded chain from our degradation pool. That last reading is our own synthesis, not the organisers'.

Table 5 gives the results. On SocialRF, squish wins on the files as distributed, after JPEG re-encoding, and within size deciles, and on its large degraded images the frozen-model contrasts favor downscaling (Sec. 6). On CommunityAI the pooled gain is confounded: every real image is a JPEG and every fake a PNG, and native size alone separates the classes with AUC 0.916. The size-matched gain on the re-encoded files does not clear zero ([$-0.002$, +0.012]). On Synthbuster squish loses by 0.001, with both fusions near ceiling. Under our degradation chains squish wins on all three sets, by 2.4 to 8.5 points.

**The second test read.** Table 6 gives both reads of the public test split. Read 1 used the crop fusion as frozen on 2026-09-23. Read 2 used the squish fusion as frozen before any squish model was scored on test. No third test evaluation was performed. Robust AUC rose by +0.062 [+0.040, +0.083] and clean AUC fell by 0.014 [0.006, 0.024]. These are complete-system differences, including training, checkpoint selection, and fusion settings; they do not isolate a frozen-model preprocessing effect. On robust AUC the squish fusion would sit between the sixth- and seventh-placed challenge entries on the open test, where the crop fusion was below all nine. We were not a challenge entry and make no ranking claim. The leading entries use far more compute, such as committees of DINOv3 models at 512 pixels on 32 A100 GPUs (Gushchin et al. 2026).

**The clean cost.** The observed test clean-AUC loss is concentrated in branch A on large images. Clean AUC of branch A on test images with a short side above 1100 pixels falls from 0.974 under crop to 0.880 under squish (366 images). Below 1100 pixels, and on branch B at every size, the change is under 1.3 points. The fake images drive it: their mean score under squish falls with size. The validation splits do not show this cost, and the test split carries no generator labels. Synthbuster, which has them, shows no single location either. The two A models also differ in weight averaging and training round, not only in input policy. We tried to remove the cost with a per-image mix of squish and grid9, weighted by the branch's predicted degradation. It lost robust AUC against squish alone ($-0.005$ [$-0.010$, $-0.001$]), so we did not spend a test read on it. Our hypothesis is the trade-off the forensic literature predicts: a clean, large generated image keeps fine traces at native resolution, and reducing it to 512 pixels discards them. This would also reconcile our result with GlobalForge (Cui et al. 2026) and TextureCrop's CLIP results (Konstantinidou, Koutlis, and Papadopoulos 2025), both measured on clean data. After degradation damages local traces, wider context may become more useful. We have not directly tested whether the detector shifts from local forensic traces to content cues.

**Public detectors.** UnivFD (Ojha, Li, and Lee 2023), a linear probe on CLIP ViT-L/14 features trained on 224-pixel crops, is at chance on NTIRE data under every presentation (robust AUC 0.49 to 0.51 on the hard split), so it says nothing either way. B-Free (Guillaro et al. 2025) is a DINOv2-B with registers that averages five 504-pixel windows taken at native resolution.[2] We scored it with its authors' pipeline (native), at its own training field of view (short side to its training size, then its crop), and under squish (Table 7). Squish changes its robust AUC on the hard split by $-0.018$ [$-0.041$, +0.004],

[2]Model size and window count from the released weights and code, https://github.com/grip-unina/B-Free.

| | SocialRF | CommunityAI | Synthbuster |
|---|---|---|---|
| n (real/fake) | 2994/3000 | 6000/6000 | 994/8996 |
| short side p50 | 1366 | 500 | 833 |
| size-label AUC | 0.694 | 0.916 | 0.519 |
| format-label AUC | 0.500 | 1.000 | 0.555 |
| crop fusion | 0.9633 | 0.9809 | 0.9995 |
| squish fusion | 0.9776 | 0.9920 | 0.9984 |
| difference | +0.014 | +0.011 | $-0.001$ |
| 95% CI | [+0.010, +0.018] | [+0.009, +0.013] | $[-0.002, -0.001]$ |
| size-matched, JPEG 95 | +0.016 | +0.005 | n/a |
| 95% CI | [+0.009, +0.024] | $[-0.002, +0.012]$ | |
| degraded (ours) | +0.085 | +0.070 | +0.024 |

Table 5: External benchmarks, separately trained crop and squish fusions. Size-matched is the size-weighted AUC within native-size deciles containing both classes (all 5,994 SocialRF images; 5,920 of 12,000 CommunityAI images). It cannot be computed on Synthbuster because no decile holds both classes (Sec. 9).

| read/system | robust | clean | AUC | TPR@1% |
|---|---|---|---|---|
| 1, crop | 0.7931 | 0.9856 | 0.9108 | 0.289 |
| 2, squish | 0.8547 | 0.9715 | 0.9211 | 0.300 |

Table 6: Complete-system comparison on `ntire_test` (2,500 images). The two reads use different trained checkpoints. Read 1's fusion weight was fitted on both validation splits ($w = 0.45$), read 2's on `ntire_val` alone.

and the change falls with image size instead of rising. Both resized presentations lower its clean AUC on every split, by 1.2 to 18.8 points, as the advice to avoid resizing predicts for a detector built on native-resolution traces. Under controlled degradations its squish deficit shrinks, from $-0.114$ with none to $-0.035$ at the heaviest level, but never crosses zero. On SocialRF under our degradation chains squish gains it 6.3 points [+5.1, +7.5]. On large degraded SocialRF images (short side at least 2,016 pixels), with B-Free's own 504-pixel windows, downscaling before cutting windows raises its robust AUC by 7.9 points [+5.3, +10.5] while coverage changes it by $-0.003$, and squish scores 0.928 against 0.804 for its native five windows. On the clean images of the same set, downscaling costs it 1.7 points and squish 3.3. B-Free therefore shows a similar preprocessing trade-off on this SocialRF subset: downscaling helps degraded inputs but costs clean AUC. Its NTIRE results show that this benefit does not hold across all settings.

## 9 Two label leaks

**AIGIBench CommunityAI.** In the evaluated copy, all 6,000 real images are JPEG files and all 6,000 fake images are PNG files, so container format separates the classes perfectly, and native short side separates them with AUC 0.916. Results on this subset should be reported after format equalisation and within size strata.

| split | native | train FOV | squish |
|---|---|---|---|
| hard-val, robust | 0.636 | 0.631 | 0.618 |
| hard-val, clean | 0.909 | 0.753 | 0.722 |
| `ntire_val`, robust | 0.735 | 0.754 | 0.744 |
| `ntire_val`, clean | 0.920 | 0.848 | 0.825 |
| SocialRF | 0.922 | 0.889 | 0.871 |
| degraded (ours) | 0.757 | 0.829 | 0.820 |
| CommunityAI | 0.891 | 0.742 | 0.703 |
| degraded (ours) | 0.683 | 0.568 | 0.539 |
| Synthbuster | 0.998 | 0.987 | 0.976 |
| degraded (ours) | 0.896 | 0.893 | 0.892 |

Table 7: B-Free AUC under three presentations. Native is the authors' own five-window test pipeline. Hard-val is `ntire_val_hard`.

**Synthbuster with resized RAISE reals.** Synthbuster ships only generated images (Bammey 2024).[3] A widely used copy of its RAISE reals, released with B-Free (Guillaro et al. 2025), was resized so that each image's area is about $1024^2$ pixels.[4] We used this copy. All 994 reals then have a short side of 833 to 835 pixels, while the fakes range from 256 to 2048. The monotone size-label AUC is 0.519 because the reals sit in the middle of the fakes' range, but the rule "short side within 833 to 835" separates the classes almost perfectly. A monotone leak check misses this. In the same benchmark, Adobe Firefly images saved with a `.png` extension are JPEG internally (Ricker, Fischer, and Quiring 2026), and 997 of our fakes are JPEG.

## 10 Limitations

Every detector we tested is a vision transformer: our two DINOv2 branches, three short CLIP runs and B-Free. The low-level and convolutional detectors that motivate the advice to avoid resizing may not share the effect, and SAFE's results

[3] The dataset record states that the RAISE-1k images are not included: https://zenodo.org/records/10066460.

[4] B-Free README, https://github.com/grip-unina/B-Free.

suggest they do not (Li et al. 2025a). UnivFD is at chance on this data. The policy was chosen on `ntire_val_hard`, so its numbers there are optimistic. The external sets and the declared test read provide evidence beyond policy selection, and their results are mixed. The decomposition uses frozen NTIRE-trained models and one held-out set, with short sides of at least 800 pixels (2,016 for B-Free). Downscaling widens each view but also changes interpolation and leaves fewer windows to average. The resizing controls cannot isolate these effects, and intervals including zero do not establish equivalence. The order test cannot tell degradation removal apart from degradation that hurts more on a smaller image, and it stops at 4x while squish downscales large images 5 to 18 times. The 512-pixel comparisons are single short runs, and they could not settle how much enlarging the window costs. Every training image has a short side of 512 pixels or less, so narrow views at native scale were never in training. The clean cost on large test images is real, unexplained, and not addressed here. Combining a whole-image view with native windows, as HiDA-Net and GLASS do (Mu et al. 2026; Han 2026), is the obvious next step, but it would need its own held-out test.

## 11 Conclusion

Input resolution alone does not describe what an image detector observes. In our experiments, presenting the whole image substantially improves robustness for frozen, crop-trained DINOv2 detectors. On degraded SocialRF images, downscaling before cropping also benefits CLIP and B-Free, while additional coverage at fixed scale contributes little. Resizing controls are consistent with a benefit from wider per-crop context, but leave interpolation and aggregation effects unresolved. Whole-image presentation also incurs clean-image losses in several settings. Detection evaluations should therefore report crop field of view, resize factors, and aggregation policy alongside input resolution, and assess preprocessing separately on clean and degraded images. Combining whole-image context with native windows is a useful next step requiring its own held-out evaluation.

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